PPE Gate: Wire-by-Wire Connection Guide

For someone who knows nothing about connections. This guide tells you, one wire at a time, what to connect where. Every wire has a number (W1, W2, ...). Do them in order, tick each box, and test at the checkpoints.

Use it together with the Hardware Build Guide. That guide has the test programs for each checkpoint (Stages 1 to 6) and the steps for building the cardboard gate. This guide is only about the connections.

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1. Read this first

The golden rules

1. **Unplug the USB cable every time you change a wire.** Plug it back in only to test.
2. **One group at a time.** Connect a group, test it, then move on. Never wire everything and then test.
3. **Never join 5V directly to GND** with a wire. That is a short circuit and can damage the Arduino.
4. **Take a photo** of your breadboard after each group, before you plug in USB. The photo helps you compare with the figures here.

Wire colours (a habit that prevents mistakes)

Colour	Use it for
Red	5V (the + rail and anything that takes power from it)
Black or blue	GND (the - rail and anything that returns to it)
Any other colour	Signal wires (the ones that go to Arduino pins 4, 5, 6, 8, 9, 10, A4, A5)

If you don't have those colours, any colour works. Just keep the red and black rule for power and ground, so you can see at a glance what each wire does.

Two kinds of jumper wire you will use

Type	Looks like	Used for
Male-to-male (M-M)	A metal pin at both ends	Arduino to breadboard, breadboard to breadboard, and into the servo's plug (it has sockets, not pins)
Male-to-female (M-F)	A pin at one end, a small socket at the other	Connecting to a part that has its own **pins** (the LCD and the sensor): the **socket end goes onto the part's pin**, and the pin end goes into the breadboard or the Arduino

Tools: nothing is needed apart from the parts. A pair of small pliers helps to straighten bent legs.

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2. Know your parts: which pin is which

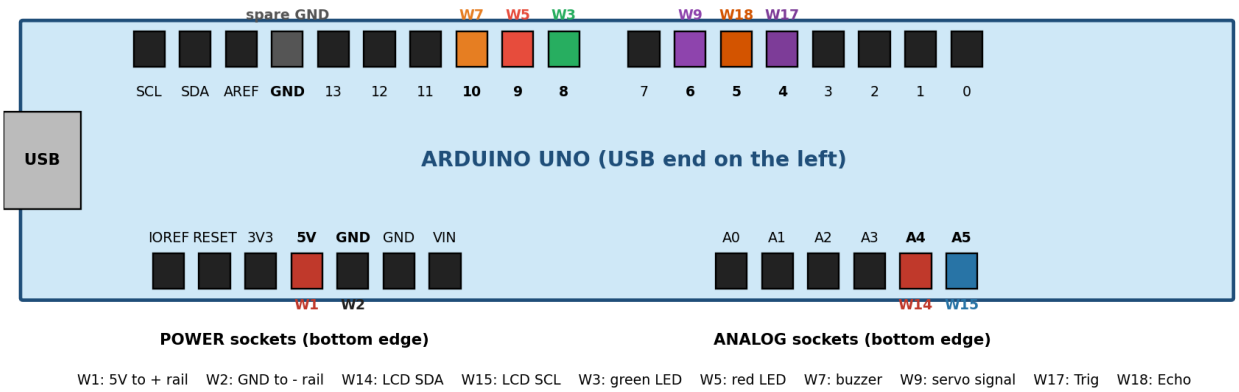
Look at each part before you connect it.

Part	What to look at	Which is which
LED (green and red)	The two legs	The **long leg is +**. The **short leg is -**. The flat edge on the rim is on the - side.
Resistor (220 ohm)	The coloured bands	Red, red, brown, gold. It has **no direction**, so either way round is fine.
Active buzzer	The top and the legs	The **longer leg is +** (a + mark is often printed on top). The shorter leg is -.
Servo (SG90)	The three-wire plug	**Brown or black = GND**, **red = 5V**, **orange or yellow = signal**.
I2C LCD	The 4 pins on the small board at the back	Read the printed labels: **GND, VCC, SDA, SCL**. The order differs between makers, so always read them.
HC-SR04 sensor	The 4 pins on the board	Read the printed labels: **VCC, Trig, Echo, GND**.

Part	What to look at	Which is which
Capacitor (470 uF)	The stripe on the side	The leg under the **stripe with a minus sign is -**. The other (usually longer) leg is **+**. Wrong way round can damage it.

3. Know the Arduino: where the sockets are

Figure B: the Arduino sockets you will use (coloured). Check the labels printed on your board.



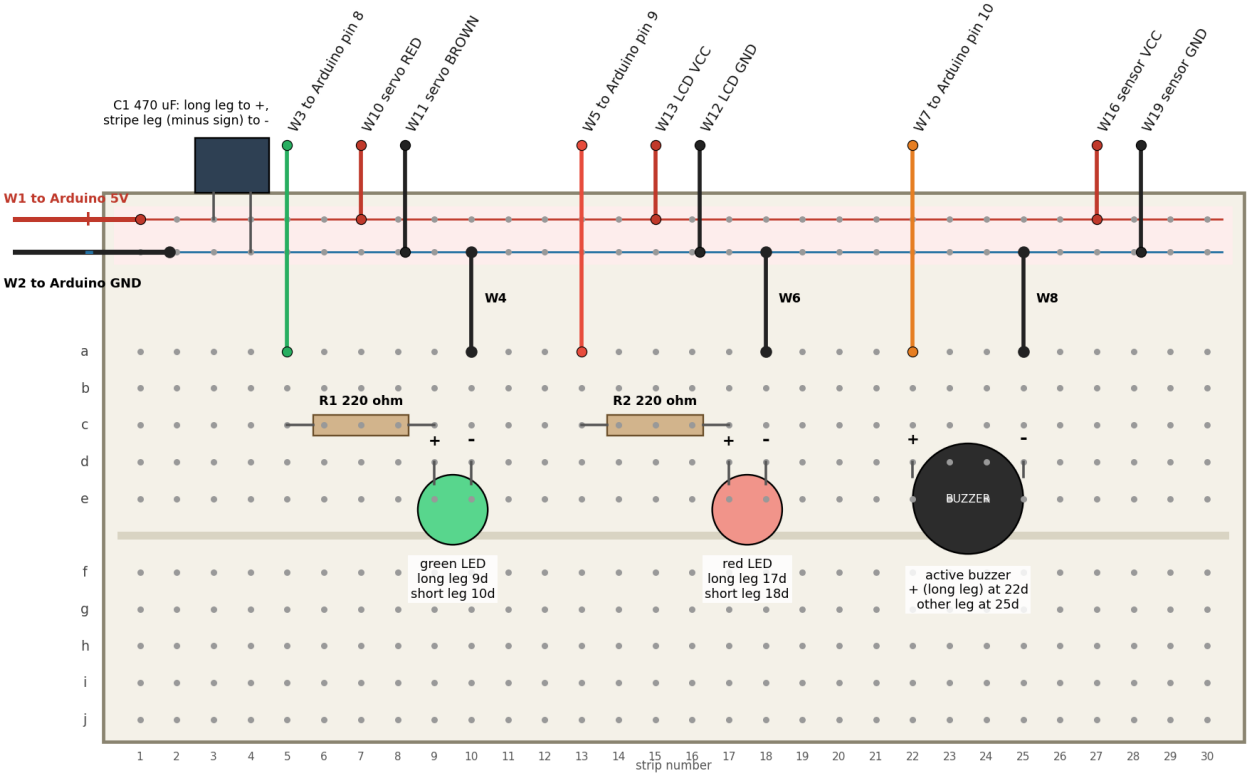
How to read it: hold the Arduino so the USB socket is on the left. Along the **top edge** are the digital sockets (0 to 13). Along the **bottom edge** are the power sockets on the left (5V, GND) and the analog sockets A0 to A5 on the right. The coloured sockets in Figure B are the ones you will use. Every socket has its name printed next to it on the board, so always confirm the label before you plug a wire in.

4. Know the breadboard: how the holes are joined

A breadboard connects parts without soldering. Some holes are joined inside the board.

- **Strips (the numbered columns, 1 to 30).** In each strip, the five holes **a, b, c, d, e** are joined together. The five holes **f, g, h, i, j** are joined together separately. The gap in the middle keeps the two groups apart.
- **Rails (the long lines along the top).** Every hole in a rail is joined to every other hole in the same rail. In this guide the top **+ rail** carries 5V and the top **- rail** carries GND.
- **What this means:** two wires or legs in the same group of five holes are electrically connected. Two parts in different strips are not.
- **Letters may run the other way on your board.** On some breadboards the letters a-e are printed from the other side. That does not matter, as long as a part's legs and its wire share the same group of five holes in a strip. In this guide I always use rows a to e.

Figure A: where everything goes on the half-size breadboard (top rails used for + and -)



Reading Figure A: the top two lines are the rails (red line +, blue line -). Below them are rows a to e, then the gap, then rows f to j. The coloured vertical lines are jumper wires. Wires that go above the board go to the Arduino (or to a part with its own pins). Wires that stop at a rail are short links.

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5. All connections at a glance

Wire	From	To	Colour suggestion	Type
W1	Arduino 5V	+ rail	Red	M-M
W2	Arduino GND	- rail	Black	M-M
W3	Arduino pin 8	Strip 5, hole 5a	Green	M-M
W4	Hole 10a	- rail	Black	M-M
W5	Arduino pin 9	Strip 13, hole 13a	Orange	M-M
W6	Hole 18a	- rail	Black	M-M
W7	Hole 22a	Arduino pin 10	Yellow	M-M
W8	Hole 25a	- rail	Black	M-M
W9	Servo orange wire	Arduino pin 6	Orange	M-M
W10	Servo red wire	+ rail	Red	M-M
W11	Servo brown wire	- rail	Black	M-M
W12	LCD GND	- rail	Black	M-F
W13	LCD VCC	+ rail	Red	M-F
W14	LCD SDA	Arduino A4	Blue	M-F
W15	LCD SCL	Arduino A5	Purple	M-F
W16	Sensor VCC	+ rail	Red	M-F
W17	Sensor Trig	Arduino pin 4	White	M-F
W18	Sensor Echo	Arduino pin 5	Grey	M-F
W19	Sensor GND	- rail	Black	M-F

Parts that sit on the breadboard (no wire number):

Part	Where
Resistor R1 (220 ohm)	Hole 5c to hole 9c
Green LED	Long leg in 9d, short leg in 10d
Resistor R2 (220 ohm)	Hole 13c to hole 17c
Red LED	Long leg in 17d, short leg in 18d
Buzzer	Long leg (+) in 22d, other leg in 25d
Capacitor C1 (470 uF)	Long leg in the + rail, striped leg in the - rail

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6. Build it one wire at a time

Step 1: Power rails (2 wires)

These two wires give every part its power and ground. Everything else depends on them.

Done	Wire	From	To
[ ]	W1 (red)	Arduino **5V** socket (bottom edge)	any hole in the **+ rail** at the far left
[ ]	W2 (black)	Arduino **GND** socket (bottom edge, next to 5V)	any hole in the **- rail** at the far left

How: push the pin end of the wire gently into the Arduino socket until it is firm, then push the other end into the rail hole. Do not force it.

Check: the + rail is the line near the top edge of the board, usually marked with red. Make sure W1 is in the + rail, and W2 is in the other rail (marked with blue or -). If you swap them, the LCD and sensor will not work and you risk damage.

Checkpoint A: plug in USB. The green power light on the Arduino stays on and nothing gets warm.

Step 2: Green LED (4 things)

The LED needs a resistor so it does not burn out, and a way back to ground.

Done	Item	From	To
[ ]	W3 (green wire)	Arduino **pin 8**	breadboard hole **5a**
[ ]	Resistor R1	hole **5c**	hole **9c**
[ ]	Green LED	**long leg** in **9d**	**short leg** in **10d**
[ ]	W4 (black wire)	hole **10a**	the **- rail**

How the electricity flows: pin 8 sends power to strip 5, the resistor carries it to strip 9, the LED's long leg takes it in, the short leg lets it out to strip 10, and W4 returns it to ground.

Check: the long leg is in strip 9 (with the resistor), and the short leg is in strip 10 (with W4). If the LED is the wrong way round, it will not light. It does not get damaged by this.

Checkpoint B: upload the LED test from the Hardware Build Guide (Stage 1). The green LED should blink.

Step 3: Red LED (4 things)

Same as the green LED, using different strips and pin 9.

Done	Item	From	To
[ ]	W5 (orange wire)	Arduino **pin 9**	breadboard hole **13a**
[ ]	Resistor R2	hole **13c**	hole **17c**
[ ]	Red LED	**long leg** in **17d**	**short leg** in **18d**
[ ]	W6 (black wire)	hole **18a**	the **- rail**

Checkpoint C: run the same LED test. Now the green LED and then the red LED should blink in turn.

Step 4: Buzzer (3 things)

Done	Item	From	To
[ ]	Buzzer	**long leg (+)** in **22d**	**other leg (-)** in **25d**
[ ]	W7 (yellow wire)	hole **22a**	Arduino **pin 10**
[ ]	W8 (black wire)	hole **25a**	the **- rail**

Note: the buzzer's legs are wider apart than normal parts. If they do not reach strips 22 and 25 neatly, bend the legs slightly outwards. The buzzer must be an **active** one. A passive one stays silent.

Checkpoint D: run the buzzer test from the Hardware Build Guide (Stage 2). You should hear a short beep every second.

Step 5: Capacitor and servo (4 things)

The capacitor smooths the power so the Arduino does not restart when the servo moves.

Done	Item	From	To
[ ]	Capacitor C1	**long leg** in the **+ rail**	**striped leg** in the **- rail**
[ ]	W9 (orange, M-M)	servo **orange** wire	Arduino **pin 6**
[ ]	W10 (red, M-M)	servo **red** wire	the **+ rail**
[ ]	W11 (black, M-M)	servo **brown** wire	the **- rail**

How to connect the servo: the servo ends in a small plastic plug with three wires (brown, red, orange), and the plug has three **sockets**. Push the **pin end of a male-to-male wire** into each socket, one wire per socket, matching the colour order of the servo wires: the brown wire's socket gets W11, the red wire's socket gets W10, and the orange wire's socket gets W9. Then connect the other ends as

in the table. The plug is a snug fit, so push gently, and wrap a little tape around the join so it cannot slip off.

Check: the colours must match: red to + rail, brown to - rail, orange to pin 6. Swapping red and brown can damage the servo.

Checkpoint E: run the servo test from the Hardware Build Guide (Stage 3). The horn swings back and forth.

Step 6: LCD (4 wires)

Done	Wire	From (LCD pin)	To
[ ]	W12 (black, M-F)	**GND**	the **- rail**
[ ]	W13 (red, M-F)	**VCC**	the **+ rail**
[ ]	W14 (blue, M-F)	**SDA**	Arduino **A4**
[ ]	W15 (purple, M-F)	**SCL**	Arduino **A5**

How: the socket end of each wire goes on the LCD's pin. Read the printed labels on the LCD's back board before connecting, because the pin order can differ.

Check: SDA goes to A4 and SCL goes to A5. If they are swapped, the LCD shows nothing.

Checkpoint F: run the LCD test from the Hardware Build Guide (Stage 4). "PPE Gate" and "Hello!" should appear. Turn the small blue contrast screw if the screen looks blank.

Step 7: Ultrasonic sensor (4 wires)

Done	Wire	From (sensor pin)	To
[ ]	W16 (red, M-F)	**VCC**	the **+ rail**
[ ]	W17 (white, M-F)	**Trig**	Arduino **pin 4**
[ ]	W18 (grey, M-F)	**Echo**	Arduino **pin 5**
[ ]	W19 (black, M-F)	**GND**	the **- rail**

Check: Trig goes to pin 4, and Echo goes to pin 5. Swapped wires give "no echo" or the number 999.

Checkpoint G: run the sensor test from the Hardware Build Guide (Stage 5). Open the Serial Monitor and move your hand in front of the sensor: the number changes.

Step 8: Final test

Upload the final sketch (Stage 6 in the Hardware Build Guide) and type the commands `C`, `G`, `R`, `H`, `V` and `I` in the Serial Monitor. Every output (LEDs, buzzer, servo, LCD) should respond. You are finished with the wiring if all of them work.

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7. Count your wires

A quick check that you have not missed anything.

What	Count
M-M jumper wires (W1 to W11)	11
M-F jumper wires (W12 to W19)	8
Resistors	2
LEDs	2
Buzzer	1
Capacitor	1
Servo, LCD, sensor, Arduino, breadboard	1 each

For the finished gate you also need about 6 more M-F wires (4 to extend the LED legs, 2 for the buzzer) to reach the front panel (see the Hardware Build Guide, Stage 7).

8. Common mistakes and how to spot them

Mistake	How it shows up	Fix
LED the wrong way round	The LED never lights	Swap the two legs.
5V wire in the - rail (or GND in the + rail)	LCD blank, sensor dead	Put W1 in the + rail and W2 in the - rail.
Buzzer legs reversed	No sound	Swap the legs.
Passive buzzer	No sound with the code	Swap it for an active buzzer.
LCD SDA and SCL swapped	Blank LCD, scanner finds nothing	SDA to A4, SCL to A5.
Sensor Trig and Echo swapped	Always 999 or "no echo"	Trig to pin 4, Echo to pin 5.
Servo red and brown swapped	Servo does not move, may get hot	Red to + rail, brown to - rail. Unplug at once if it feels hot.
Capacitor the wrong way round	The Arduino resets, the capacitor may get warm	Long leg to +, striped leg to -.
A part not in the same group of five holes as its wire	That part does nothing	Check the wire and the leg share the same strip, rows a-e (or f-j).
Loose wire	Works, then stops	Press every wire firmly into its hole, and tape the female ends.
No common ground	Strange behaviour	Every part's GND must go to the - rail, and the - rail to Arduino GND (W2).

9. Photo checklist

Before you power the finished board, take a photo and compare:

Check	Done
W1 is red and goes to the + rail; W2 is black and goes to the - rail	[ ]
Both LEDs: long leg on the resistor side, short leg towards the wire to the - rail	[ ]
Buzzer's long leg is in strip 22	[ ]
Capacitor's striped leg is in the - rail	[ ]
Servo: orange to pin 6, red to + rail, brown to - rail	[ ]
LCD: SDA on A4, SCL on A5	[ ]
Sensor: Trig on pin 4, Echo on pin 5	[ ]
No two bare metal parts touching each other	[ ]

If you send me the photo with a short description of what is not working, I can help you find the mistake.